

AI/ML Hackathon - Ministry of Panchayati Raj (Geospatial Intelligence Challenge)

Powered by: **Geo-Intel Lab, IITTNiF**

Problem Statement 1: Feature Extraction from Drone Images

**AI-Driven Drone-Based System for Real-Time Air Quality
Mapping and Pollution Detection**



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Supported by

**Geospatial Intelligence and Applications Laboratory,
IIT Tirupati Navavishkar I-Hub Foundation**

1. Title of the Idea

AI-Driven Drone-Based System for Real-Time Air Quality Mapping and Pollution Detection

2. Team/Individual Name & Affiliation

Team

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3. Problem Statement Addressed

Air pollution is a growing concern in urban environments, leading to severe health issues and environmental degradation. Conventional monitoring systems depend on fixed air quality sensors that are expensive, sparsely distributed, and incapable of capturing real-time spatial variations. This results in incomplete data, delayed responses, and ineffective pollution control strategies. Pollution hotspots such as traffic congestion zones, construction sites, and industrial areas remain under-monitored. There is a need for an intelligent, scalable, and cost-efficient system that can dynamically assess environmental conditions across large areas. By combining drone-based aerial imaging with artificial intelligence, it is possible to analyze land-use patterns and detect pollution-prone regions. Such a system can enhance monitoring accuracy, enable real-time insights, and support data-driven decision-making for environmental sustainability.

4. Proposed Solution

The proposed solution is an AI-powered drone surveillance system that performs real-time environmental monitoring and air quality estimation using computer vision and deep learning.

Drones capture high-resolution aerial images across different regions. These images are processed using a trained AI model to detect and classify environmental features such as roads, buildings, vegetation, water bodies, and industrial zones. These features are strongly correlated with pollution sources (e.g., traffic emissions, construction dust, industrial pollutants).

The system integrates image-based feature extraction with geospatial mapping to generate pollution heatmaps. Advanced versions can incorporate real-time data streams and combine them with external datasets (weather, traffic density, satellite data) to improve prediction accuracy.

A lightweight deep learning model (CNN-based) ensures efficient processing and deployment. The output is visualized through an interactive dashboard that highlights pollution-prone areas, enabling authorities to take preventive actions.

This solution is scalable, cost-effective, and adaptable to smart city ecosystems. It transforms raw aerial imagery into actionable environmental intelligence, enabling proactive air quality management.

5. Uniqueness and Innovation

This solution stands out due to its multi-dimensional innovation:

- **Drone + AI Integration:** Combines aerial surveillance with deep learning for dynamic environmental monitoring.
- **Indirect Pollution Estimation:** Uses land-use features as proxies instead of relying solely on physical sensors.
- **Real-Time Spatial Intelligence:** Provides high-resolution pollution mapping across large areas.
- **Cost Efficiency:** Eliminates need for dense sensor infrastructure.
- **Scalable Framework:** Easily deployable across cities, rural areas, and industrial zones.
- **Future-Ready Design:** Can integrate advanced models (CNNs, Vision Transformers) and real-time analytics.

Unlike traditional systems, this approach transforms visual data into environmental insights, making it highly adaptive and innovative.

6. Technology Stack / Methodology

The system is built using:

- **Programming:** Python
- **AI Frameworks:** TensorFlow, Keras
- **Image Processing:** OpenCV
- **Data Handling:** NumPy
- **Deployment (future):** Cloud (AWS/GCP), IoT integration

Methodology:

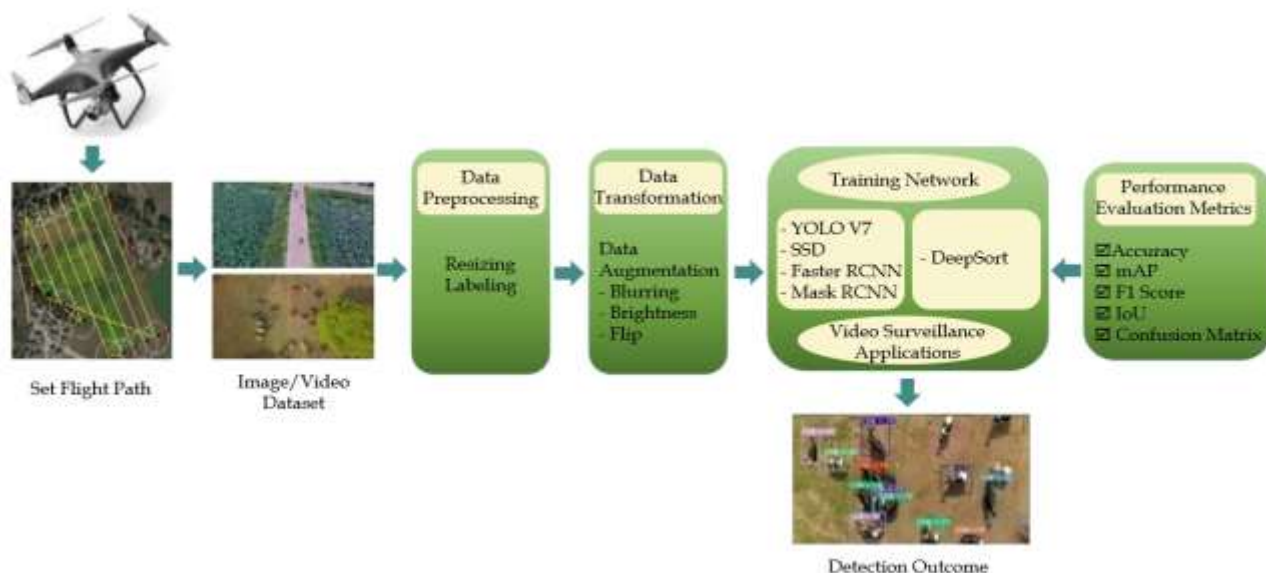
1. Data collection via drones
2. Image preprocessing (resize, normalization)
3. Feature extraction using CNN model
4. Classification of environmental elements

5. Mapping predictions to pollution indicators

6. Visualization using dashboards / GIS

The current prototype uses a simplified neural network model (trained on MNIST for demonstration), but the architecture is designed to scale to real-world aerial datasets.

Architecture



7. Expected Impact (Environmental, Social, Economic, etc.)

The system enables real-time identification of pollution hotspots, improving environmental monitoring and decision-making. It supports smarter urban planning, reduces health risks, and lowers monitoring costs. Governments can implement targeted interventions, while communities benefit from cleaner air and improved living conditions.

8. Implementation Plan / Roadmap

Phase 1 (0–2 months): Prototype development and validation

Phase 2 (2–5 months): Dataset collection (drone images) and model enhancement

Phase 3 (5–8 months): Integration with GIS and dashboard development

Phase 4 (8–12 months): Pilot deployment in selected region

Phase 5: Scaling and optimization

9. Required Resources

- GPU-enabled system
- Drone hardware / datasets
- AI/ML tools (TensorFlow, OpenCV)
- Cloud infrastructure
- Domain expertise (AI + environmental science)

10. Scalability and Sustainability

The system can be deployed across cities using drone networks and cloud platforms. It supports continuous monitoring with minimal infrastructure. Sustainable design ensures adaptability to future technologies and environmental needs.

11. Stakeholders and Collaborations (if any)

- Government environmental agencies
- Smart city authorities
- Research institutions
- Drone service providers
- AI/tech companies

12. Current Stage of Development

Stage: Prototype

A functional prototype is being developed demonstrating image processing and classification capabilities. While currently tested on sample datasets, the system is ready to be extended to real-world drone imagery and environmental applications.

13. Supporting Materials (Optional)

GitHub Repository:

<https://github.com/Narender-9/IIT-TIRUPATI>